

Lab Class: Physics

Specific Charge (e/m) of electron

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1 Introduction

The aim of this experiment is to determine the specific charge e/m of the electron, which represents the ratio of the electron's charge to its mass. The value of e/m is crucial for understanding atomic physics, as it allows the calculation of the electron mass m_e when combined with the elementary charge e , determined later in the Millikan experiment.

2 Theory

When an electron moves through a magnetic field, it experiences the *Lorentz force* $\vec{F}_L$ given by:

$$\vec{F}_L = -e(\vec{v}_e \times \vec{B}) \quad (1)$$

with e being elementary charge ($1.602 \cdot 10^{-19}C$), $\vec{v}_e$ being the electron velocity and $\vec{B}$ being the magnetic flux density.

If the electron moves perpendicular to $\vec{B}$, the *Lorentz force* acts as a centripetal force, causing the electron to follow a circular path of radius r . The *centrifugal force* $\vec{F}_z$ balances the Lorentz force:

$$\vec{F}_z = \frac{m_e \cdot v_e^2}{r} = \vec{F}_L = e \cdot v_e \cdot \vec{B}_z$$

Equating the two forces gives:

$$\frac{m_e \cdot v_e}{r} = e \cdot B_z \quad (2)$$

Helmholtz coils are a pair of identical circular coils separated by a distance equal to their radius R . When the same current flows through both coils in the same direction, they produce a nearly uniform magnetic field in the central region between them. The magnetic flux density at the center ($z = 0$) is:

$$B_z = I \cdot \left(\frac{4}{5}\right)^{\frac{3}{2}} \cdot \frac{N \cdot \mu_0}{R} \quad (3)$$

with I being the supplied current for both coils (it's the same for both), N being the number of windings per coil and μ_0 the permeability of free space ($4 \cdot \pi \cdot 10^{-7} \frac{N}{A^2}$).

Free electrons which pass through a voltage difference of an acceleration voltage U_B , have an energy of:

$$W = e \cdot U_B$$

These electrons convert electrical potential energy to kinetic energy.

$$W = e \cdot U_B = W_{kin} = \frac{1}{2} \cdot m_e \cdot v_e^2$$

The velocity of these electrons can be computed as:

$$v_e = \sqrt{\frac{2 \cdot e \cdot U_B}{m_e}} \quad (4)$$

Substituting this v_e to the force balance equation:

$$\frac{m_e}{r} \cdot \sqrt{\frac{2 \cdot e \cdot U_B}{m_e}} = e \cdot B_z \Rightarrow \frac{e}{m_e} = \frac{2 \cdot U_B}{B_z^2 \cdot r^2} \quad (5)$$

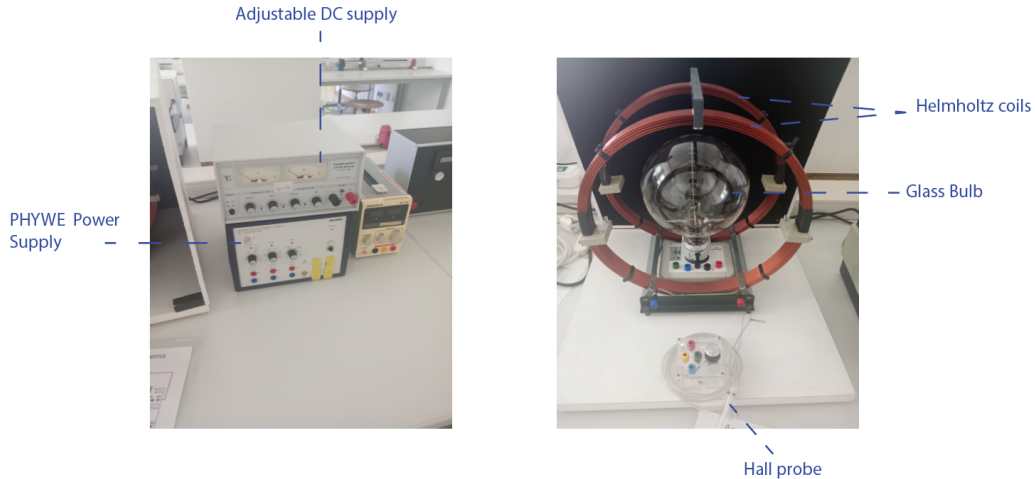
3 Experimental Setup

The experiment was set up in a laboratory provided by the *Hochschule München University of Applied Sciences*, which is used for a variety of thermodynamics experiments.

For this experiment, the Helmholtz coils mentioned above were used. The coils were connected in series to ensure that the same current flows through them. A Hall probe was used to check the homogeneity of the magnetic field of the Helmholtz coils. As a Filament ray tube, a glass bulb filled with low-pressured gas, was used. The glass makes the path of an electron beam visible through a faint glow, and the lowered gas allows the electrons to move freely. This bulb contains an electron gun in the lower part. The tube is located in the center of a pair of Helmholtz coils that generate a magnetic flux density which can be considered homogeneous. The orbital radius can be determined with the grid provided in the glass bulb with a spacing of 2 cm, the zero point of which is in the exit opening of the electron gun.

Different power supplies were also used:

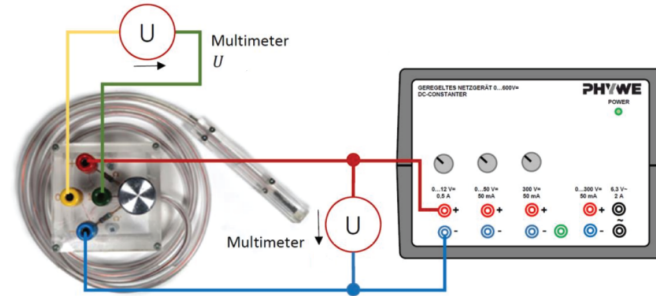
- High-voltage supply for electron acceleration
- Low-voltage supply for heating the cathode
- Adjustable DC supply for Helmholtz coil current



(6)

4 Experimental Produce

The first experiment was carried out to calibrate the magnetic field between the Helmholtz coils. This was done using the Hall probe by moving it through the field with distinct deviations. The circuit was set as follows:



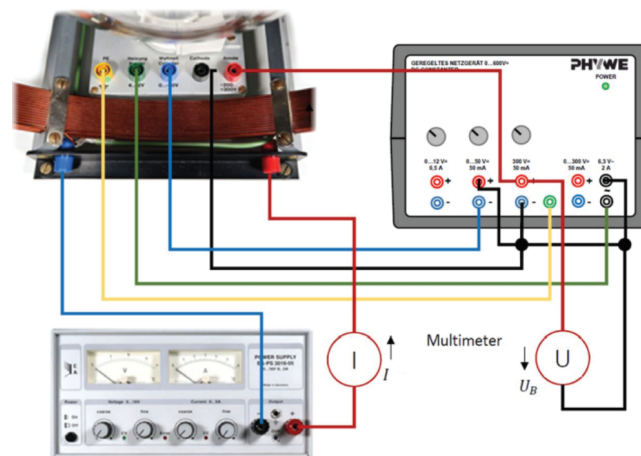
(7)

The voltage source was set to apply 10V to the Hall probe and a current of 3A was supplied to the coils. The Hall probe was initially placed at the center of the coils and moved by steps of 10mm in the z -direction in reference to the coils up to 140mm. The output voltage at each position was read using a multimeter and recorded. This was done for both the positive and the negative z -direction. The output voltage at each position was recorded as follows:

	0mm	10mm	20mm	30mm	40mm	50mm	60mm	70mm	80mm	90mm	100mm	110mm	120mm	130mm	140mm
U in the -ve z-axis(V)	128.8	128.3	127.8	127.3	126.8	125.8	123.8	121.4	117.3	111.2	103.2	92.4	79.1	62.9	46
U in the +ve z-axis(V)	124.9	124.5	124.5	124.4	123.7	122.5	120.3	117.1	112.2	105.4	95.5	83.7	68.2	50.8	34.1

(8)

The second experiment was setup as follows:



(9)

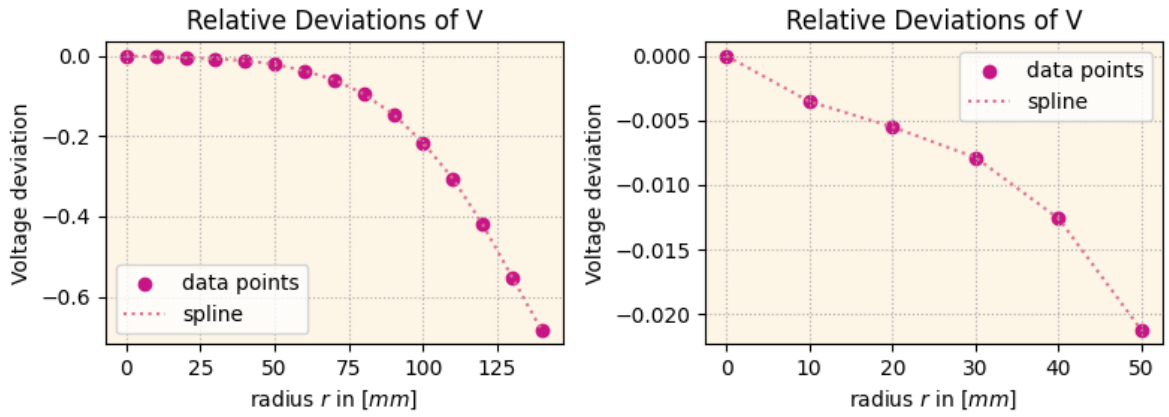
An acceleration voltage of 300V was supplied. By covering the bulb with a box, the natural light in the room could be dimmed around the bulb and a faint circular glow could be observed. The voltage applied to the Wehnelt cylinder (metal electrode surrounding the cathode in the

electron gun) was adjusted until the beam was concentrated enough. Next, the coil current was adjusted for the path of electrons to have a radius of $40mm$. This current was found to be $1.9A$. This process was repeated to increase accuracy. The same $1.9A$ current was found to yield the $40mm$ radius. Since the current for both times was the same the average current will be $1.9A$ again.

Using the data obtained from the first experiment, it was possible to calculate the relative deviations at different positions:

$$\frac{\Delta U(r)}{U(r=0)} = \frac{U(r) - U(r=0)}{U(r=0)}$$

The result were as follows:



(10)

	0mm	10mm	20mm	30mm	40mm	50mm	60mm	70mm	80mm	90mm	100mm	110mm	120mm	130mm	140mm
Deviations	0	-0.0035	-0.0055	-0.0079	-0.0126	-0.0213	-0.0378	-0.0599	-0.0954	-0.1462	-0.2168	-0.3059	-0.4194	-0.5518	-0.6843

(11)

As seen from the table, the deviation of the magnetic flux density at path of $40mm$ radius can be calculated as:

$$1 - 0.0126 = 0.9874 = 98.74\%$$

Then using equation (3), the magnetic flux density at center was calculated. Which yielded a value of $B_z = 1.4362 \cdot 10^{-3}T$. At radius of $40mm$ can then be given by:

$$1.4362 \cdot 10^{-3}T \cdot 0.9874 = 1.4181 \cdot 10^{-3}T$$

This is then used to calculate the specific charge of the electrons by placing it in equation (5). The voltage $300V$ was used as an input voltage U_B . This gives a value of $1.8647 \cdot 10^{11} \frac{C}{kg}$.

In all the measurements in these experiments, the resistances present in the wires and multi meters should be acknowledged. So the values used during the calculations could have a slight deviation from the literature values. Additionally the accuracy of the human eye when identifying the radius of the beam lacks precision and could influence this deviation.

The exact values are described in the table below:

Parameters	Measured uncertainties
Acceleration voltage	$\Delta U_B = 1.8V$
Current	$\Delta I = 0.088A$
Radius of beam	$\Delta r = 4 \cdot 10^{-3}m$
Output volatge(r=0)	$\Delta U_0 = 0.3V$
Output volatge(r=40)	$\Delta U_{40} = 0.3V$

(12)

Parameters	Calculation of uncertainty	Value
Deviation	$\Delta(\frac{\Delta U_c}{U_{r0}}) = \sqrt{(\frac{U_{40}}{U_0} \cdot \Delta U_0)^2 + (\frac{1}{U_0} \cdot U_{40})^2}$	$\Delta(\frac{U_c}{U_{r0}}) = 0.0049$
Magnetic flux density	$\Delta B_z = \sqrt{((\frac{\Delta U_c}{U_{r0}}) \cdot (\frac{4}{5})^{\frac{3}{2}} \cdot \frac{N \cdot \mu_0}{r} \cdot \Delta I)^2 + (-\frac{\Delta U_c}{U_{r0}}) \cdot (\frac{4}{5})^{\frac{3}{2}} \cdot \frac{I \cdot N \cdot \mu_0}{r^2} \cdot \Delta r)^2 + ((\frac{4}{5})^{\frac{3}{2}} \cdot \frac{I \cdot N \cdot \mu_0}{r} \cdot \Delta(\frac{\Delta U_c}{U_{r0}}))^2}$	$\Delta B_z = 0.5769 \cdot 10^{-3}T$
Specific charge of electron	$\Delta(e/m_e) = \sqrt{(\frac{2}{B_z^2 \cdot r^2} \cdot \Delta U_B)^2 + (-\frac{4 \cdot U_B}{B_z^2 \cdot r^2} \cdot \Delta B_z)^2 + (-\frac{4 \cdot U_B}{B_z^2 \cdot r^2} \cdot \Delta r)^2}$	$\Delta(e/m_e) = 1.5623 \cdot 10^{11} \frac{C}{kg}$

(13)

The calculated value of the specific charge ($1.8647 \cdot 10^{11} \frac{C}{kg}$) has a deviation of 6.02% from the literature value $1.7588 \cdot 10^{11} \frac{C}{kg}$.

5 Summary

In this experiment, the specific charge of the electron (e/m) was determined using a filament ray tube and Helmholtz coils. By accelerating electrons with a known voltage (U_B) and deflecting them in a uniform magnetic field (B_z) in a circular path radius (r). The experimental results are faithful to theory to an unexpectedly high extent, considering the limited experimental precision and the small magnitudes of the calculations.

Literature

Preparation Sheet, provided by Prof. Dr. Matthias Gramich, Prof. Dr. Daniel Förster, [Specific charge e/m of the electron (SLE)]